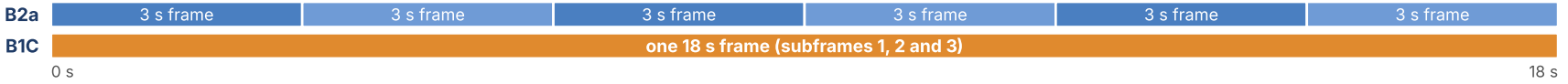
BeiDou navigation messages: B-CNAV2 (B2a) and B-CNAV1 (B1C)

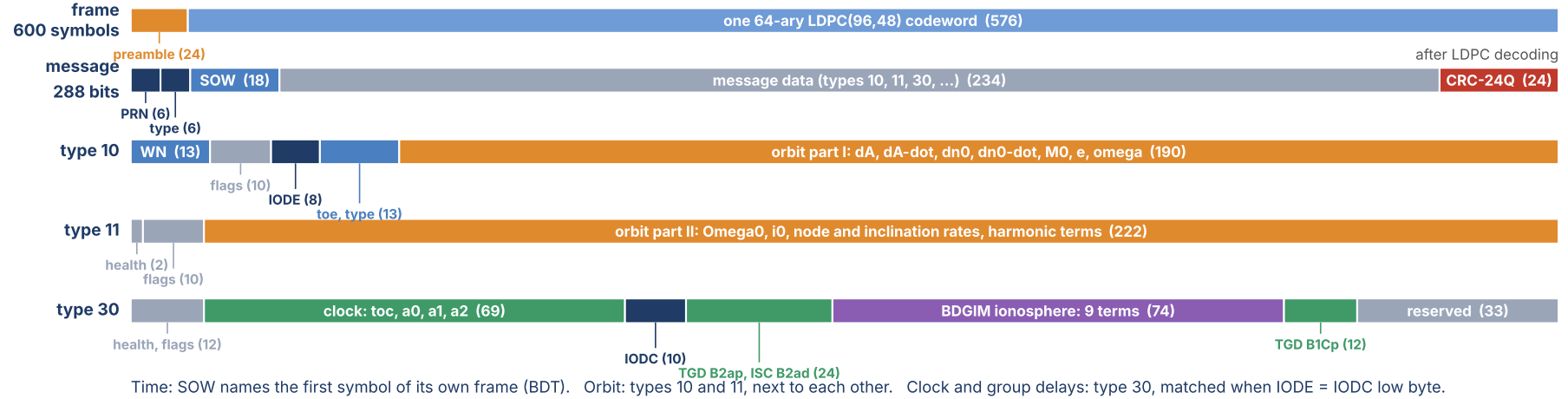
Bar width is proportional to the number of bits or symbols. Field sizes come from the decoder code.

A. The same 18 seconds on the two signals



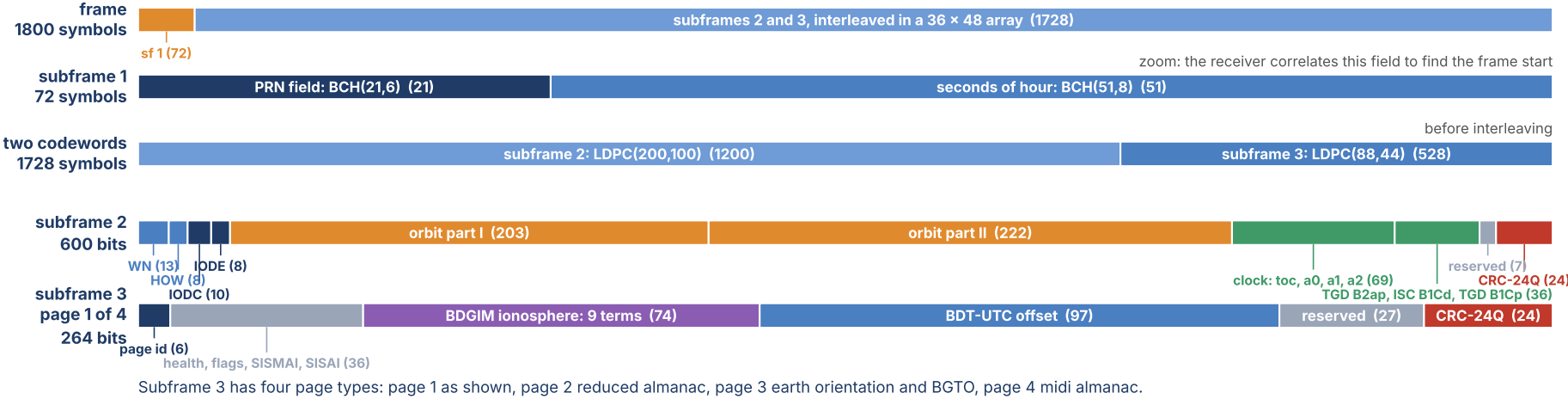
B2a: six frames of 3 s. The message types rotate. A full ephemeris needs type 10 + type 11 + type 30 (30 s of tracking gave all three on every satellite of the real recording).
B1C: one frame of 18 s. Subframe 2 holds the whole ephemeris and the clock, so one good frame is enough.

B. B-CNAV2 on B2a: 600 symbols at 200 sps = 3 s



Time: SOW names the first symbol of its own frame (BDT). Orbit: types 10 and 11, next to each other. Clock and group delays: type 30, matched when IODE = IODC low byte.

C. B-CNAV1 on B1C: 1800 symbols at 100 sps = 18 s, no preamble



Subframe 3 has four page types: page 1 as shown, page 2 reduced almanac, page 3 earth orientation and BGTO, page 4 midi almanac.
Time: frame start = HOW x 3600 + SOH x 18 s (BDT), with WN from subframe 2. Ephemeris, clock, group delays and week: subframe 2 only.
Receiver steps: find the start with the PRN field and the SOH field, undo the interleaver, LDPC-decode both codewords, check the CRC-24Q of each subframe.

time · orbit · clock, group delay · identity, issue numbers · ionosphere · health, integrity · CRC-24Q